

Review Exercise 2

2. Express the following numbers in scientific notation:

(i) 0.000567

$$0.000567 = 5.67 \times 10^{-4}$$

(ii) 734

$$734 = 7.34 \times 10^2$$

(iii) 0.33×10^3

$$\begin{aligned} 0.33 \times 10^3 &= 3.3 \times 10^{-1} \times 10^3 \\ &= 3.3 \times 10^{-1+3} \\ &= 3.3 \times 10^2 \end{aligned}$$

3. Express the following numbers in ordinary notation:

(i) 2.6×10^3

$$2.6 \times 10^3 = 2600$$

(ii) 8.974×10^{-4}

$$8.974 \times 10^{-4} = 0.0008974$$

(iii) 6×10^{-6}

$$6 \times 10^{-6} = 0.000006$$

4. Express each of the following in logarithmic form:

(i) $3^7 = 2187$

$$\begin{aligned} 3^7 &= 2187 \\ \log_3 2187 &= 7 \end{aligned}$$

(ii) $a^b = c$

$$\begin{aligned} a^b &= c \\ \log_a c &= b \end{aligned}$$

(iii) $12^2 = 144$

$$\begin{aligned} 12^2 &= 144 \\ \log_{12} 144 &= 2 \end{aligned}$$

5. Express each of the following in exponential form:

(i) $\log_4 8 = x$

$$\begin{aligned} \log_4 8 &= x \\ 4^x &= 8 \end{aligned}$$

(ii) $\log_9 729 = 3$

$$\begin{aligned} \log_9 729 &= 3 \\ 9^3 &= 729 \end{aligned}$$

(iii) $\log_4 1024 = 5$

$$\begin{aligned} \log_4 1024 &= 5 \\ 4^5 &= 1024 \end{aligned}$$

6. Find the value of x in the following:

(i) $\log_9 x = 0.5$

$$\begin{aligned} \log_9 x &= 0.5 \\ 9^{0.5} &= x \\ 9^{\frac{5}{10}} &= x \end{aligned}$$

$$\begin{aligned}
 9^{\frac{1}{2}} &= x \\
 (3^2)^{\frac{1}{2}} &= x \\
 3 &= x \\
 \mathbf{x} &= \mathbf{3}
 \end{aligned}$$

$$\text{(ii)} \left(\frac{1}{9}\right)^{3x} = 27$$

$$\begin{aligned}
 \left(\frac{1}{9}\right)^{3x} &= 27 \\
 \left(\frac{1}{3^2}\right)^{3x} &= 3^3 \\
 (3^{-2})^{3x} &= 3^3 \\
 3^{-6x} &= 3^3 \\
 \Rightarrow -6x &= 3 \\
 x &= \frac{3}{-6} \\
 \mathbf{x} &= \mathbf{\frac{1}{-2}}
 \end{aligned}$$

$$\text{(iii)} \left(\frac{1}{32}\right)^{2x} = 64$$

$$\begin{aligned}
 \left(\frac{1}{32}\right)^{2x} &= 64 \\
 \left(\frac{1}{2^5}\right)^{2x} &= 2^6 \\
 (2^{-5})^{2x} &= 2^6 \\
 2^{-10x} &= 2^6 \\
 \Rightarrow -10x &= 6 \\
 x &= \frac{6}{-10} \\
 \mathbf{x} &= \mathbf{-\frac{3}{5}}
 \end{aligned}$$

7. Write the following as a single logarithm:

$$\text{(i)} 7 \log x - 3 \log y^2$$

$$\begin{aligned}
 &7 \log x - 3 \log y^2 \\
 &= \log x^7 - \log (y^2)^3 \\
 &= \log x^7 - \log y^6 \\
 &= \log \frac{x^7}{y^6}
 \end{aligned}$$

$$\text{(ii)} 3 \log 4 - \log 32$$

$$\begin{aligned}
 &3 \log 4 - \log 32 \\
 &= \log 4^3 - \log 32 \\
 &= \log \frac{4^3}{32}
 \end{aligned}$$

$$\text{(iii)} \frac{1}{3} (\log_5 8 + \log_5 27) - \log_5 3$$

$$\begin{aligned}
 &\frac{1}{3} (\log_5 8 + \log_5 27) - \log_5 3 \\
 &= \frac{1}{3} \log_5 2^3 + \frac{1}{3} \log_5 3^3 - \log_5 3 \\
 &= \log_5 (2^3)^{\frac{1}{3}} + \log_5 (3^3)^{\frac{1}{3}} - \log_5 3
 \end{aligned}$$

$$= \log_5 2 + \log_5 3 - \log_5 3$$

$$= \log_5 2$$

8. Expand the following using laws of logarithms:

(i) $\log(xyz^6)$

$$\log(xyz^6)$$

$$= \log x + \log y + \log z^6$$

$$= \log x + \log y + 6 \log z$$

(ii) $\log_3 \sqrt[6]{m^5 n^3}$

$$\log_3 \sqrt[6]{m^5 n^3}$$

$$= \log_3 (m^5 n^3)^{\frac{1}{6}}$$

$$= \frac{1}{6} [\log_3 m^5 n^3]$$

$$= \frac{1}{6} [\log_3 m^5 + \log_3 n^3]$$

$$= \frac{1}{6} [5 \log_3 m + 3 \log_3 n]$$

(iii) $\log \sqrt{8x^3}$

$$\log \sqrt{8x^3}$$

$$= \log (2^3 x^3)^{\frac{1}{2}}$$

$$= \frac{1}{2} [\log 2^3 x^3]$$

$$= \frac{1}{2} [\log 2^3 + \log x^3]$$

$$= \frac{1}{2} [3 \log 2 + 3 \log x]$$

$$= \frac{3}{2} [\log 2 + \log x]$$

9. Find the values of the following with the help of logarithm table:

(i) $\sqrt[3]{68.24}$

Let

$$x = \sqrt[3]{68.24}$$

$$\log x = \log \sqrt[3]{68.24}$$

$$\log x = \log (68.24)^{\frac{1}{3}}$$

$$\log x = \frac{1}{3} \log (68.24)$$

$$\log x = \frac{1}{3} (1.8340)$$

$$\log x = 0.6113$$

$$\text{Antilog}(\log x) = \text{Anti log}(0.6113)$$

$$x = \mathbf{4.086}$$

(ii) 319.8×3.543

Let

$$x = 319.8 \times 3.543$$

$$\log x = \log(319.8 \times 3.543)$$

$$\log x = \log 319.8 + \log 3.543$$

$$\log x = 2.5049 + 0.5494$$

$$\log x = 3.0543$$

$$\text{Antilog}(\log x) = \text{Anti log}(3.0543)$$

$$x = \mathbf{1133}$$

(iii) $\frac{36.12 \times 750.9}{113.2 \times 9.98}$

Let

$$x = \frac{36.12 \times 750.9}{113.2 \times 9.98}$$

$$\log x = \log \left[\frac{36.12 \times 750.9}{113.2 \times 9.98} \right]$$

$$\log x = \log 36.12 + \log 750.9 - \log 113.2 - \log 9.98$$

$$\log x = 1.5577 + 2.8756 - 2.0538 - 0.9991$$

$$\log x = 1.3804$$

$$\text{Antilog}(\log x) = \text{Anti log}(1.3804)$$

$$x = \mathbf{24.01}$$

10. In the year 2016, the population of a city was 22 millions and was growing at a rate of 2.5% per year. The function $p(t) = 22(1.025)^t$ gives the population in millions, t years after 2016. Use the model to determine in which year the population will reach 35 millions. Round the answer to the nearest year.

Initial population in 2016 = 22 millions

Population = $p(t)$ = 35 millions

Year when population will reach 35 millions = ?

Since

$$p(t) = 22(1.025)^t$$

$$35 = 22(1.025)^t$$

$$\log 35 = \log 22(1.025)^t$$

$$\log 35 = \log 22 + \log(1.025)^t$$

$$\log 35 - \log 22 = t \log 1.025$$

$$1.5441 - 1.3424 = t(0.0107)$$

$$\frac{0.2017}{0.0107} = t$$

$$18.85 = t$$

$$t \approx 19 \text{ years}$$

Now

$$\text{Year when population will be 35 million} = 2016 + 19$$

$$= \mathbf{2035}$$